Michael Purpura

Network Engineer III (Automation and DevOps · Python · CI/CD · Platform Developer)

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Summary

Network engineer with 13 years in carrier networking at Charter Communications across multiple teams: turning up customer circuits, handling tier four escalations for BGP, MPLS, QoS, and hardware outages, and serving as the team's documentation lead for about 10 years. Automated standards rollouts across thousands of Juniper, Cisco, and Nokia routers with self-taught Python, then joined Charter's DevOps team as a lead developer on an in-house network inventory and security password management system. Uses AI and LLM orchestration to build tools, including a home lab AI hub running local LLMs (Ollama) for agent experiments and a Raspberry Pi cluster manager with SSH automation and SNMP monitoring.

Skills

Automation
Python, Paramiko (SSH), Flask, SQL (MariaDB, PostgreSQL), custom reporting, Splunk, network automation, configuration management (standards, rollouts, audits), web form building (JavaScript, HTML), Visual Basic

DevOps & AI
CI/CD pipelines, Git, GitLab (merge requests, code review), production deployments and release verification, Linux servers, Docker (beginner), HashiCorp Vault (front-end use); AI / artificial intelligence, LLM orchestration, Amazon Q

Routing & Networking
BGP, MPLS, IS-IS, LDP, VPLS, L2VPN, E-LAN, Carrier Ethernet, cell backhaul, WAN, QoS, TACACS+, SNMP

Platforms & Vendors
Juniper MX and QFX, Nokia 7750; Cisco, Harmonic, RAD, Adva, Ciena, AudioCodes, Adtran; vendor management; EXFO circuit testing (RFC 2544, Y.1731)

Documentation
Confluence (space admin), Jira workflow building, documentation leadership, technical standards

Experience

Network Engineer III (DevOps Team)

October 2023 – June 2026
Charter Communications / Spectrum Enterprise, Madison, WI

- Earned a place on a dedicated DevOps team through automation work; was a lead developer on an in-house network inventory and security password management system for Charter's national enterprise footprint (hundreds of thousands of devices).
- Built the platform's inventory management system (MariaDB, later PostgreSQL), fed by live SSH discovery through PE routers and used by its security modules; wrote daily SQL for custom reports, statistics, and IP audits; documented the team's queries.
- Wrote multi-vendor Python/Paramiko modules to audit configurations and rotate device passwords over SSH across Juniper, Cisco, Nokia, Harmonic, RAD, Adva, Ciena, AudioCodes, and Adtran, with credentials in HashiCorp Vault.
- Shipped code through GitLab merge requests and CI/CD pipelines to production.

Network Engineer III
September 2013 – October 2023
Charter Communications / Spectrum Enterprise, Madison, WI

- Served as a tier four escalation point for the NOC, troubleshooting customer router outages, including BGP, MPLS, QoS, and hardware failures; covered on-call rotations.
- Provisioned and turned up thousands of enterprise circuits, mostly customer internet and voice, plus Type 2 (off-net) circuits, cell backhaul, Carrier Ethernet, and MPLS L2VPN with LDP for MEF E-Line and E-LAN services; used VRFs and L3VPN in rare cases; performed turn-up testing with EXFO.
- Selected to travel for key regional work, including serving as the engineer sent to support the network transition when Charter took over the Time Warner region; wrote the Cisco-to-Juniper configuration standards after the merger.
- Self-taught in Python through hands-on automation; used it to manage configuration standards, rollouts, and audits across thousands of routers and other network devices.
- Ran an automated pre-check, change, and post-check loop with diff checks on every router for potentially region-wide service-impacting changes, producing pre and post documentation for each device.
- Automated MOP (method of procedure) creation for the engineering team: tools that connected to devices over SSH, parsed site variables, and generated complete procedures for multi-hundred-site migrations instead of engineers writing MOPs by hand.
- Launched a WebEx bot that pinged hundreds of routers on a schedule and alerted the team to outages.
- Delivered Splunk reports from TACACS+ and ISE command logs that gave management measurable engineering metrics: commands, router commits, and hours worked per engineer, by device and maintenance window.
- Posted maintenance window summaries to WebEx, giving the day shift a clean handoff with device-level accountability.
- Served as the team's documentation lead for about 10 years: Confluence space admin and Jira workflow builder; traveled to learn new products and standards and turned them into team documentation.

Education

North Central Technical College
December 2012

Associate of Applied Science, IT Network Specialist, Unified Communications, GPA 3.8
Technical Certificate, IT Website Hardware and Software Administration

Certifications Held

- Cisco CCNP Routing and Switching
- Juniper JNCIA
- Nokia NRS I
- MEF-CECP